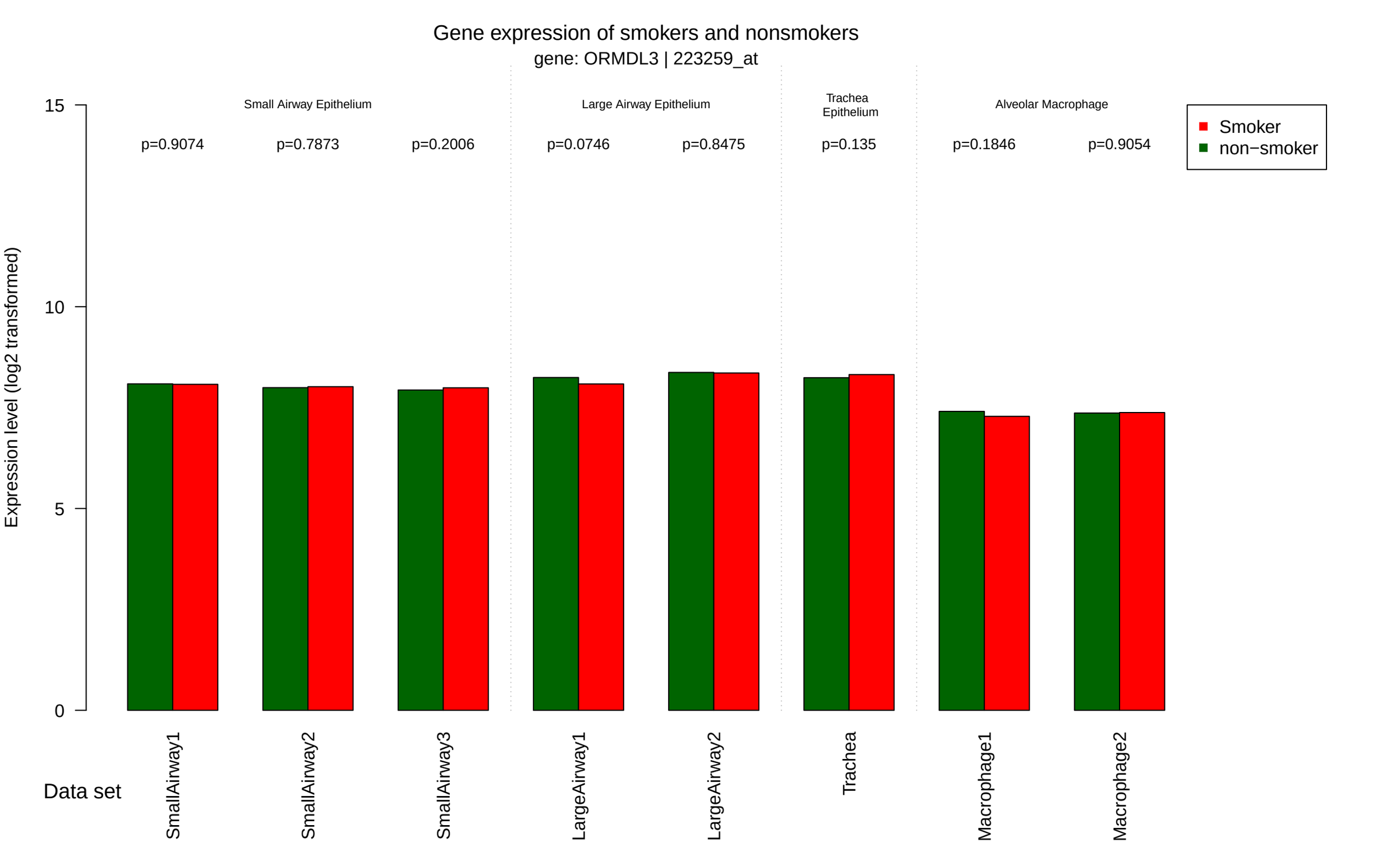
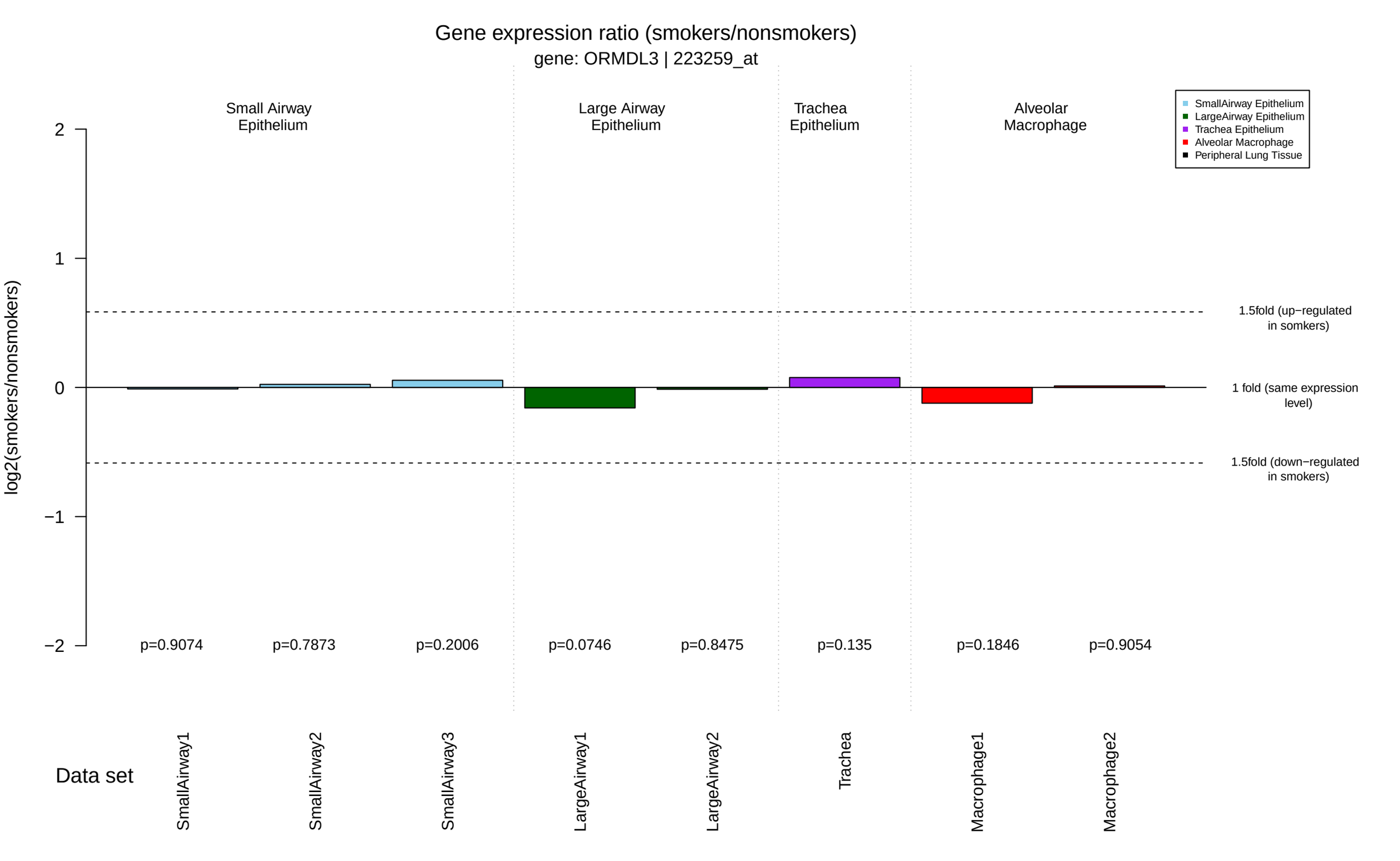
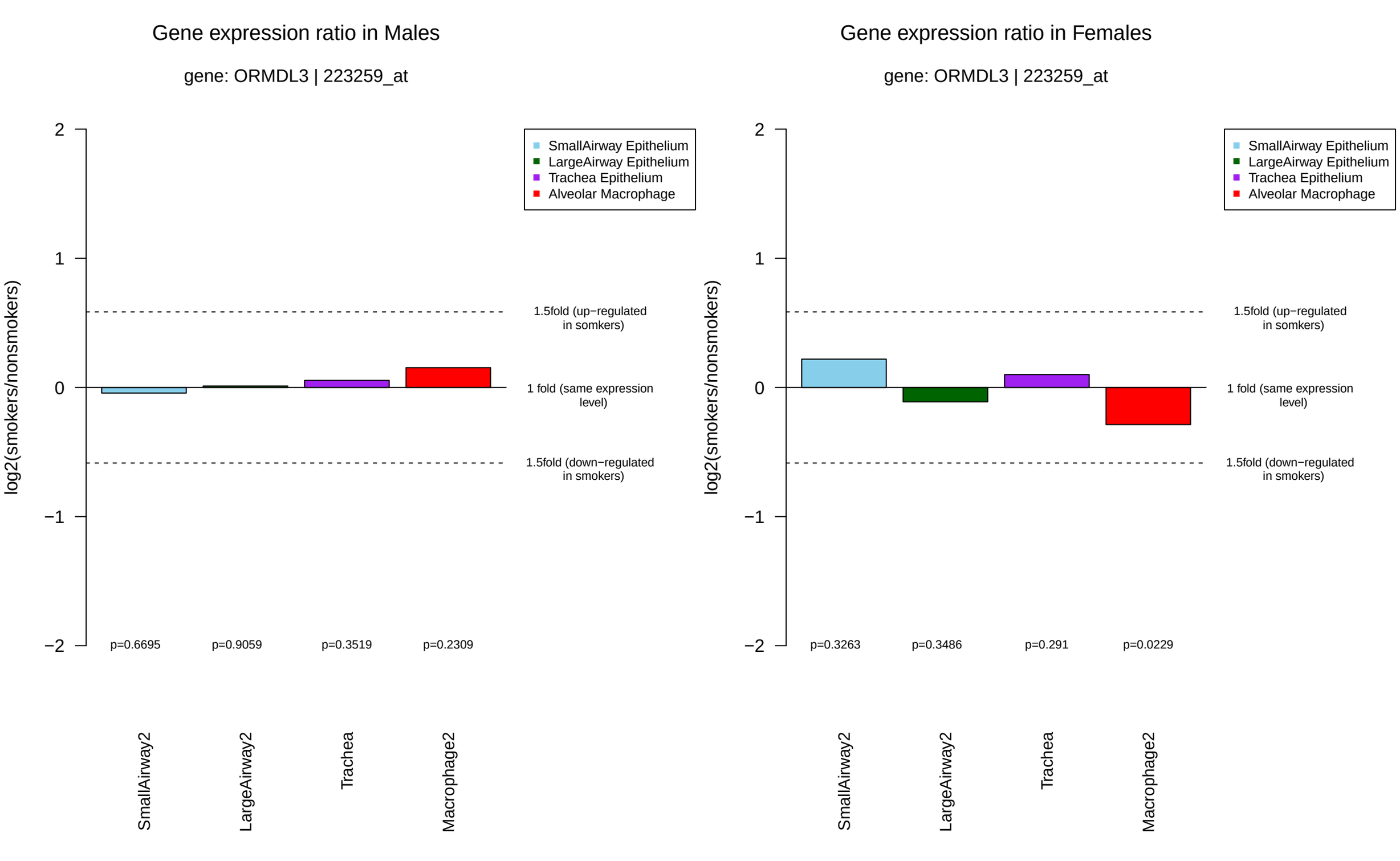
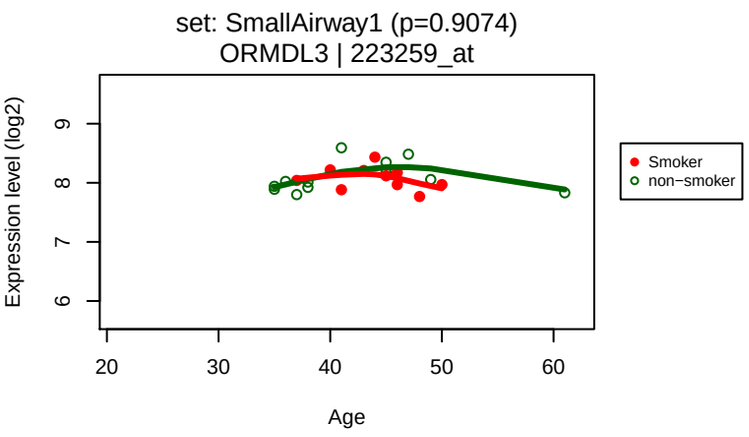
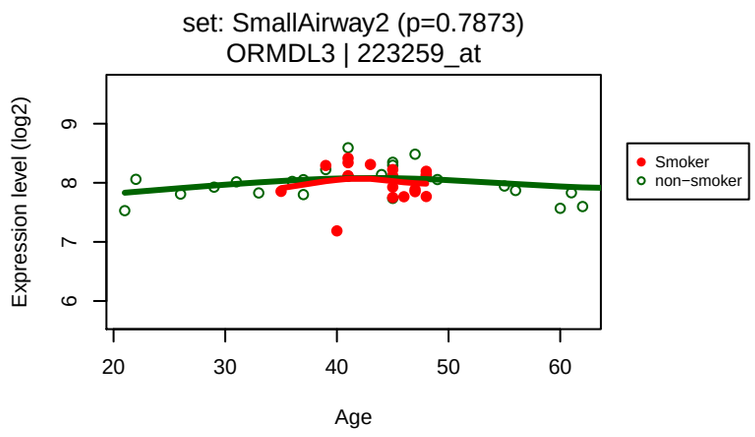
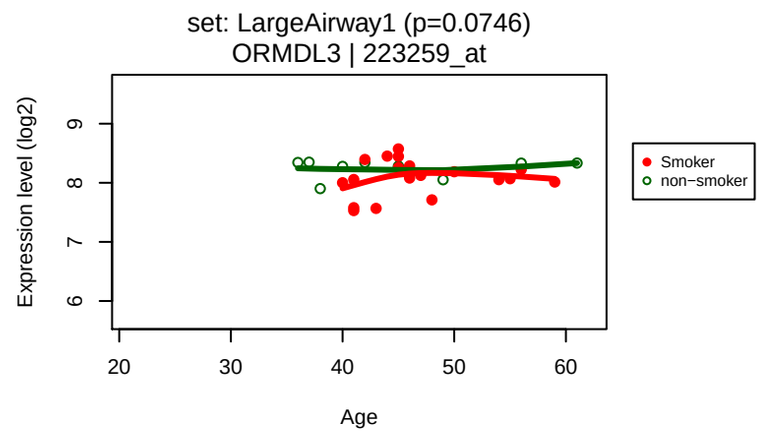

Gene expression of smokers and nonsmokers
gene: ORMDL3 | 223259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9074
p=0.7873
p=0.2006
p=0.0746
p=0.8475
p=0.135
p=0.1846
p=0.9054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

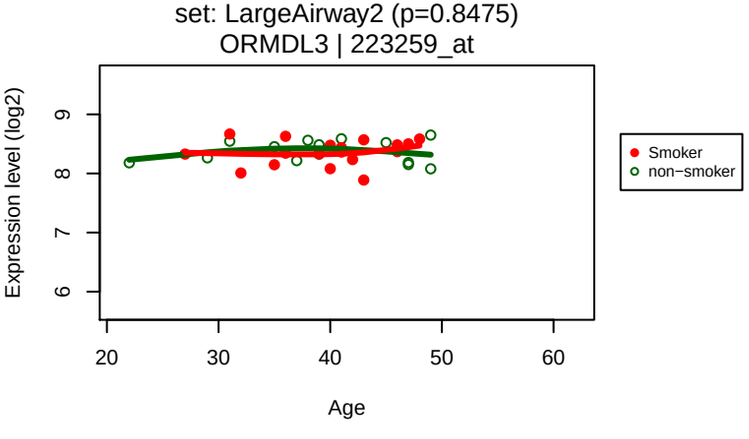

Gene expression ratio (smokers/nonsmokers)
gene: ORMDL3 | 223259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9074
p=0.7873
p=0.2006
p=0.0746
p=0.8475
p=0.135
p=0.1846
p=0.9054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

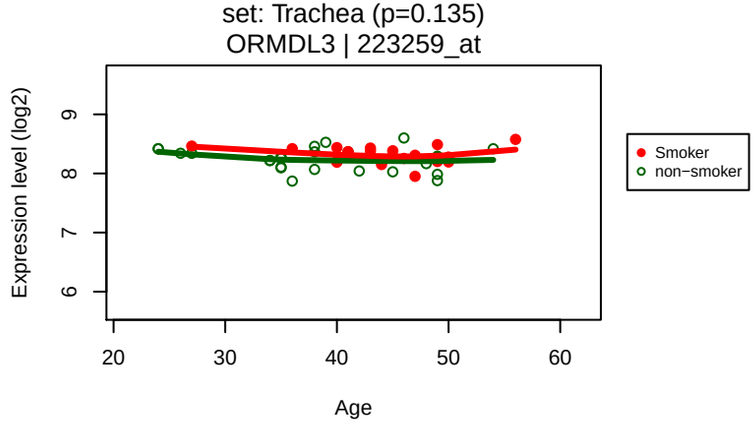

Gene expression ratio in Males
gene: ORMDL3 | 223259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6695
p=0.9059
p=0.3519
p=0.2309
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ORMDL3 | 223259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3263
p=0.3486
p=0.291
p=0.0229
SmallAirway2
LargeAirway2
Trachea
Macrophage2

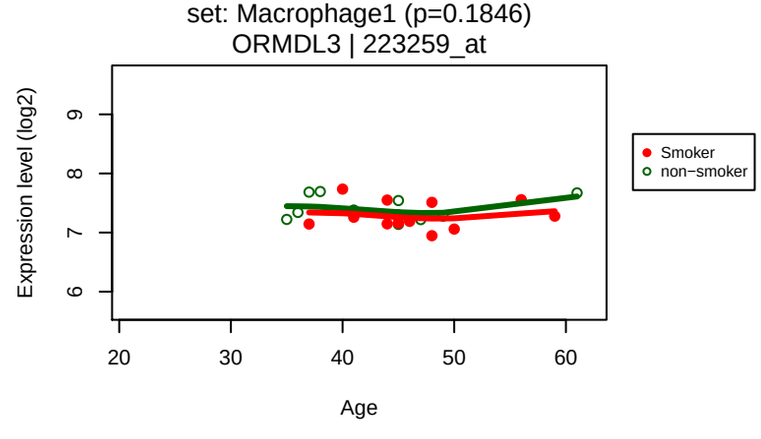

set: SmallAirway1 (p=0.9074)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker

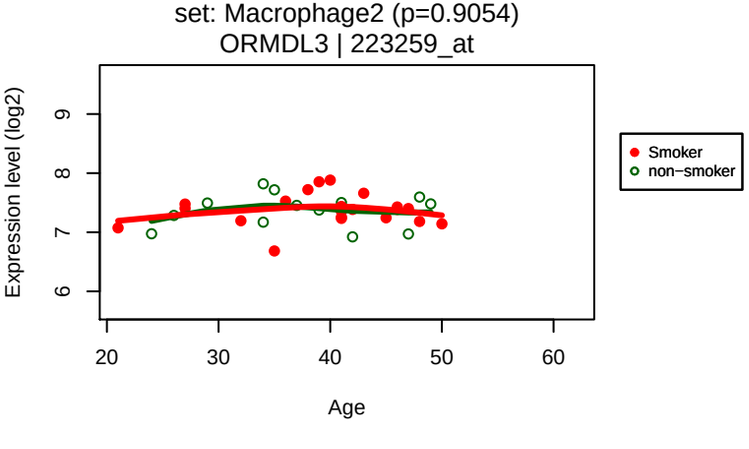

set: SmallAirway2 (p=0.7873)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker

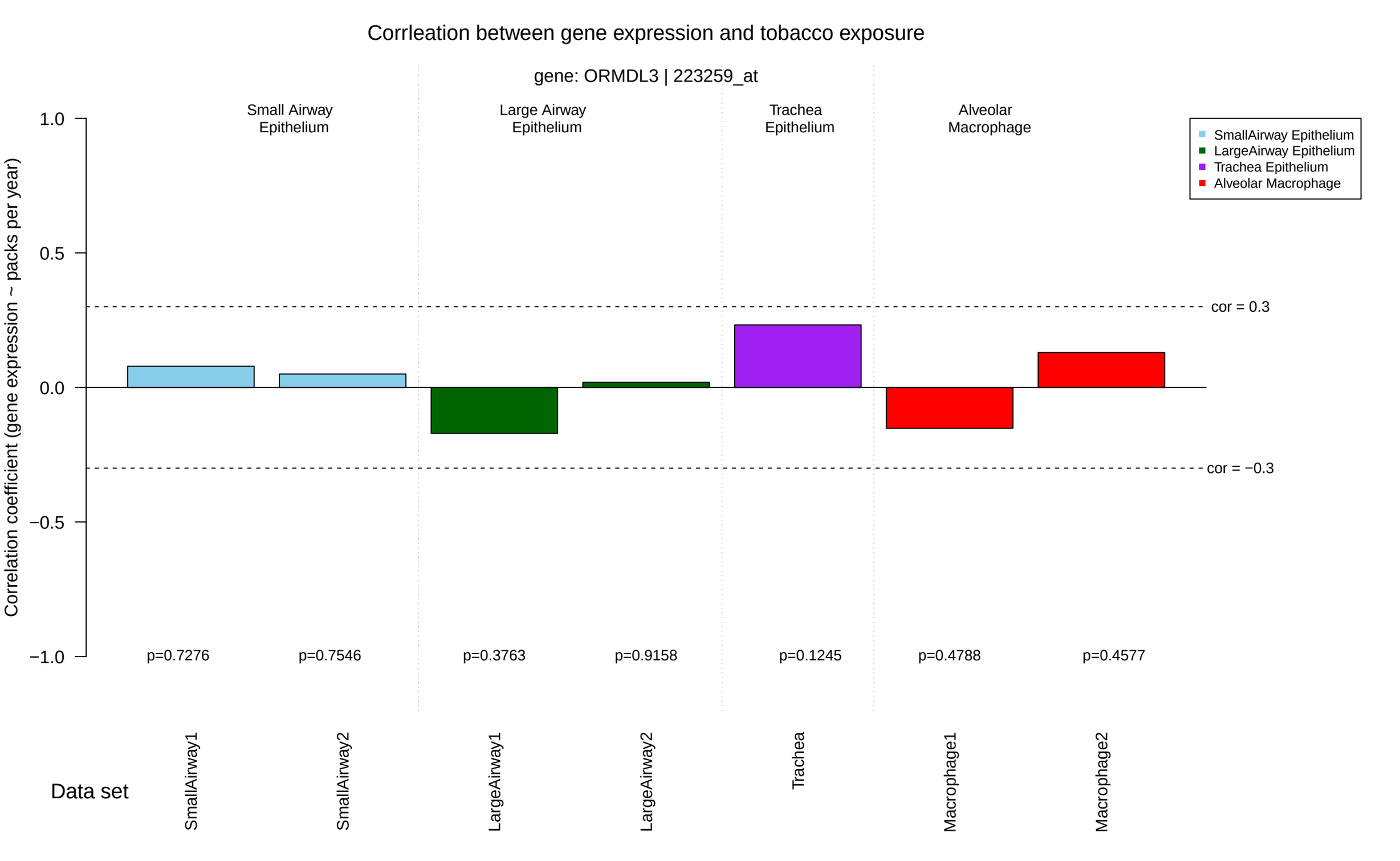

set: LargeAirway1 (p=0.0746)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8475)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.135)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1846)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9054)
ORMDL3 | 223259_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ORMDL3 | 223259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7276
p=0.7546
p=0.3763
p=0.9158
p=0.1245
p=0.4788
p=0.4577
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2